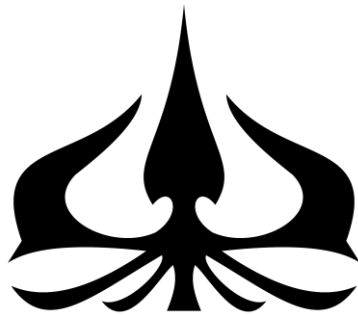


Nama: Rizal Gusti Wirawan NIM: 064102400031	 UNIVERSITAS TRISAKTI	MODUL 9 (V.1.0) Nama Dosen: Syandra Sari, M.Kom
Hari/Tanggal: Senin, 11 Juni 2026	Praktikum Manajemen Data dan Informasi	Nama Asisten Labratorium: 1. Almakius Felix Bariq Hekopung - 065002300016 2. Dewanto Maulana Sukarno Putra - 065002300004

Perintah atau Command SQL Shell “JOIN Antar Tabel”

1. Teori Singkat

Perintah ****JOIN**** dalam SQL Shell digunakan untuk menggabungkan data dari dua atau lebih tabel berdasarkan kolom yang memiliki hubungan. JOIN memungkinkan pengguna mengambil data yang terhubung antar tabel tanpa perlu menyimpan informasi yang berulang. Ada beberapa jenis JOIN dalam SQL, termasuk ****INNER JOIN****, ****LEFT JOIN****, ****RIGHT JOIN****, dan ****FULL JOIN****, yang masing-masing menentukan bagaimana data dari tabel-tabel tersebut digabungkan. Misalnya, ****INNER JOIN**** hanya mengambil data yang memiliki kecocokan di kedua tabel, sedangkan ****LEFT JOIN**** akan menampilkan semua data dari tabel kiri dan hanya data yang cocok dari tabel kanan.

Penggunaan JOIN sangat penting dalam manajemen basis data relasional, terutama untuk menghindari redundansi dan meningkatkan efisiensi dalam pengambilan data. Misalnya, dalam sistem akademik, tabel ****Mahasiswa**** dapat digabungkan dengan tabel ****Nilai**** menggunakan ****JOIN**** berdasarkan kolom ****ID_Mahasiswa**** untuk menampilkan nama mahasiswa beserta nilai mereka dalam satu hasil query. Dengan ****JOIN****, analisis data menjadi lebih mudah dilakukan, memungkinkan penyajian informasi yang lebih terstruktur dan bermakna.

2. Alat dan Bahan Hardware :

Laptop atau PC **Software**
:
PostgreSQL



3. Elemen Kompetensi

1.1 Ikuti perintah soal

1.1.1 Menampilkan phone_number, hire_date dan department_name dengan menggabungkan table employees dan table departments.

```
hr=> SELECT phone_number, hire_date, department_name FROM employees INNER JOIN departments ON employees.department_id= departments.department_id;
```

phone_number	hire_date	department_name
515.123.4567	1987-06-17	Executive
515.123.4568	1989-09-21	Executive
515.123.4569	1993-01-13	Executive
590.423.4567	1990-01-03	IT
590.423.4568	1991-05-21	IT
590.423.4569	1997-06-25	IT
590.423.4560	1998-02-05	IT
590.423.5567	1999-02-07	IT
515.124.4569	1994-08-17	Finance
515.124.4169	1994-08-16	Finance
515.124.4269	1997-09-28	Finance
515.124.4369	1997-09-30	Finance
515.124.4469	1998-03-07	Finance
515.124.4567	1999-12-07	Finance
515.127.4561	1994-12-07	Purchasing
515.127.4562	1995-05-18	Purchasing
515.127.4563	1997-12-24	Purchasing
515.127.4564	1997-07-24	Purchasing
515.127.4565	1998-11-15	Purchasing
515.127.4566	1999-08-10	Purchasing
650.123.1224	1996-07-18	Shipping
650.123.2234	1997-04-10	Shipping
650.123.3234	1995-05-01	Shipping
650.123.4234	1997-10-10	Shipping
650.123.5234	1999-11-16	Shipping
011.44.1344.429268	1996-10-01	Sales
011.44.1344.467268	1997-01-05	Sales
011.44.1344.429278	1997-03-10	Sales
011.44.1344.619268	1999-10-15	Sales
011.44.1344.429018	2000-01-29	Sales
011.44.1344.129268	1997-01-30	Sales
011.44.1344.345268	1997-03-24	Sales
011.44.1344.478968	1997-08-20	Sales
011.44.1344.498718	1998-03-30	Sales
011.44.1344.987668	1998-12-09	Sales
011.44.1344.486508	1999-11-23	Sales
011.44.1345.429268	1996-01-30	Sales
011.44.1345.929268	1996-03-04	Sales
011.44.1345.829268	1996-08-01	Sales
011.44.1345.729268	1997-03-10	Sales

```
515.123.8080 | 1994-06-07 | Accounting
515.123.8181 | 1994-06-07 | Accounting
515.123.7867 | 1997-12-14 | Shipping
(87 rows)
```

1.1.2 Menampilkan department_id, location_id, start_date, dan end_date dengan menggabungkan table departments dan job_history

```
hr=> SELECT department_name, location_id, start_date, end_date FROM departments INNER JOIN job_history USING (department_id);
```

department_name	location_id	start_date	end_date
Marketing	1800	1996-02-17	1999-12-19
Shipping	1500	1999-01-01	1999-12-31
Shipping	1500	1998-03-24	1999-12-31
IT	1400	1993-01-13	1998-07-24
Sales	2500	1999-01-01	1999-12-31
Sales	2500	1998-03-24	1998-12-31
Executive	1700	1994-07-01	1998-12-31
Executive	1700	1987-09-17	1993-06-17
Accounting	1700	1993-10-28	1997-03-15
Accounting	1700	1989-09-21	1993-10-27

1.1.3 Menampilkan seluruh data id department pada tabel department dan menampilkan start date yang ada di tabel job_history dengan department_id di job_history >20

```
hr=> SELECT departments.department_id, start_date FROM departments LEFT JOIN job_history ON departments.department_id= job_history.department_id WHERE job_history.department_id > 20;
```

department_id	start_date
50	1999-01-01
50	1998-03-24
60	1993-01-13
80	1999-01-01
80	1998-03-24
90	1994-07-01
90	1987-09-17
110	1993-10-28
110	1989-09-21



3.3.4 Menampilkan nama depan dan nama belakang dari table employees serta id lokasi dari tabel departments dimana nama depannya harus berawal dari huruf A.

```
hr=> SELECT first_name, last_name, location_id FROM employees INNER JOIN departments ON employees.department_id= departments.department_id AND first_name LI
KE 'A%';
```

first_name	last_name	location_id
Alexander	Hunold	1400
Alexander	Khoo	1700
Adam	Fripp	1500
Alberto	Errazuriz	2500
Allan	McEwen	2500
Amit	Banda	2500
Alyssa	Hutton	2500
Alexis	Bull	1500
Anthony	Cabrio	1500
Alana	Walsh	1500
Alis	ghina	1500

(11 rows)

3.3.5 Menampilkan nama depan dari tabel employees serta nama departemen dari tabel departments dimana gaji dari karyawan tersebut lebih besar dari 15000.

```
hr=> SELECT first_name, department_name FROM employees INNER JOIN departments ON employees.manager_id= departments.manager_id WHERE salary > 15000;
```

first_name	department_name
Alis	Executive
Lex	Executive
Neena	Executive

(3 rows)

3.3.6 Menampilkan semua location id yang ada di tabel departments dan menampilkan kota yang ada di tabel location.

```
hr=> SELECT departments.location_id, city FROM departments LEFT JOIN locations ON departments.location_id= locations.location_id;
```

location_id	city
1700	Seattle
1800	Toronto
1700	Seattle
2400	London
1500	South San Francisco
1400	Southlake
2700	Munich
2500	Oxford
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1700	Seattle
1800	Toronto

(27 rows)

3.3.7 Menampilkan nama depan dari tabel employees dan menampilkan semua nama departemen yang ada di tabel departemen dengan tidak berulang.



```
hr=> SELECT DISTINCT first_name, department_name FROM employees RIGHT JOIN departments ON employees.manager_id= departments.manager_id;
```

first_name	department_name
	IT Engineering
	Government Sales
Alis	Executive
	Construction
John	Finance
Karen	Purchasing
Anthony	Shipping
David	IT
Pat	Marketing
	Benefits
	Operations
	Contracting
Matthew	Executive
Nandita	Shipping
	Recruiting
Gerald	Executive
Sigal	Purchasing
	Administration
	Retail Sales
Julia	Shipping
	Corporate Tax
Guy	Purchasing
Den	Executive
Neena	Executive
Bruce	IT
	IT Helpdesk
Adam	Executive
Luis	Finance
	Manufacturing
Alberto	Executive
Karen	Executive
Shelli	Purchasing
	Human Resources
Ismael	Finance
	Shareholder Services
Michael	Executive
	Payroll
Christopher	Sales
Peter	Sales
	Treasury

3.3.8 Menampilkan location_id dari tabel departments dan menampilkan seluruh kota yang ada di tabel locations dimana status provinsinya kosong.

```
hr=> SELECT * FROM locations;
```

location_id	street_address	postal_code	city	state_province	country_id
1000	1297 Via Cola di Rie	00989	Roma		IT
1100	93891 Calle della Testa	10934	Venice		IT
1200	2017 Shinjuku-ku	1689	Tokyo	Tokyo Prefecture	JP
1400	2014 Jabberwocky Rd	26192	Southlake	Texas	US
1500	2011 Interiors Blvd	99236	South San Francisco	California	US
1600	2007 Zagora St	50090	South Brunswick	New Jersey	US
1700	2004 Charade Rd	98199	Seattle	Washington	US
1800	147 Spadina Ave	M5V 2L7	Toronto	Ontario	CA
1900	6092 Boxwood St	V5W 2T2	Whitehorse	Yukon	CA
2000	40-5-12 Laogianggen	190518	Beijing		CN
2100	1298 Vileparle (E)	400231	Bombay	Maharashtra	IN
2200	12-98 Victoria Street	2901	Sydney	New South Wales	AU
2300	198 Clementi North	540198	Singapore		SG
2500	Magdalen Centre, The Oxford Science Park	OX9 9ZB	Oxford	Oxford	UK
2600	9702 Chester Road	09629850293	Stretford	Manchester	UK
2700	Schwanthalerstr. 7031	80925	Munich	Bavaria	DE
2800	Rua Frei Caneca 1360	01307-002	Sao Paulo	Sao Paulo	BR
2900	20 Rue des Corps-Saints	1730	Geneva	Geneve	CH
3000	Murtenstrasse 921	3095	Bern	BE	CH
3100	Pieter Breughelstraat 837	3029SK	Utrecht		NL
3200	Mariano Escobedo 9991	11932	Mexico City	Distrito Federal	MX
2400	8204 Arthur St	ABC294	London		UK

(22 rows)

```
hr=> SELECT departments.location_id, city FROM departments RIGHT JOIN locations ON departments.location_id= locations.location_id WHERE state_province IS NULL;
```

location_id	city
2400	London
	Singapore
	Roma
	Venice
	Beijing

(5 rows)

TUGAS

- Menampilkan first_name, email, salary, start_date dengan menggabungkan table employees dan table job_history

```
hr=> SELECT first_name, email, salary, start_date FROM employees INNER JOIN job_history USING (employee_id);
```

first_name	email	salary	start_date
Neena	NKOCHHAR	17000.00	1993-10-28
Neena	NKOCHHAR	17000.00	1989-09-21
Lex	LDEHMAN	17000.00	1993-01-13
Den	DRAPHEAL	11000.00	1998-03-24
Payam	PKAUFILIN	7900.00	1999-01-01
Jonathon	JTAYLOR	8600.00	1999-01-01
Jonathon	JTAYLOR	8600.00	1998-03-24
Jennifer	JWHALEN	4400.00	1994-07-01
Jennifer	JWHALEN	4400.00	1987-09-17
Michael	MHARTSTE	13000.00	1996-02-17

(10 rows)

- Menampilkan seluruh data id job, job_title pada tabel jobs dan menampilkan start_date yang ada di tabel job_history.



```
hr=> SELECT jobs.job_id, job_title, start_date FROM jobs LEFT JOIN job_history ON jobs.job_id = job_history.job_id;
```

job_id	job_title	start_date
AD_PRES	President	
AD_VP	Administration Vice President	
AD_ASST	Administration Assistant	1987-09-17
FI_MGR	Finance Manager	
FI_ACCOUNT	Accountant	
AC_MGR	Accounting Manager	1993-10-28
AC_ACCOUNT	Public Accountant	1994-07-01
AC_ACCOUNT	Public Accountant	1989-09-21
SA_MAN	Sales Manager	1999-01-01
SA_REP	Sales Representative	1998-03-24
PU_MAN	Purchasing Manager	
PU_CLERK	Purchasing Clerk	
ST_MAN	Stock Manager	
ST_CLERK	Stock Clerk	1999-01-01
ST_CLERK	Stock Clerk	1998-03-24
SH_CLERK	Shipping Clerk	
IT_PROG	Programmer	1993-01-13
MK_MAN	Marketing Manager	
MK_REP	Marketing Representative	1996-02-17
HR_REP	Human Resources Representative	
PR_REP	Public Relations Representative	

(21 rows)

3. Menampilkan nama depan, nama belakang dari tabel employees dan menampilkan seluruh data id lokasi dari tabel departments dimana nama belakang dari karyawannya berakhiran huruf a.

```
hr=> SELECT first_name, last_name, location_id FROM employees INNER JOIN departments ON employees.department_id = departments.department_id AND last_name LI
```

```
KE 'A';
```

first_name	last_name	location_id
------------	-----------	-------------

(0 rows)

4. Menampilkan semua id job yang ada di table job_history dan mengambil nama job yang ada di tabel jobs dimana minimal gaji untuk job tersebut lebih besar sama dengan 4500.

```
hr=> SELECT job_history.job_id, job_title FROM job_history LEFT JOIN jobs ON job_history.job_id = jobs.job_id WHERE min_salary > 4500;
```

job_id	job_title
AC_MGR	Accounting Manager
SA_MAN	Sales Manager
SA_REP	Sales Representative

(3 rows)

5. Menampilkan nama depan dari tabel employees dan tanggal dimulai kerjanya karyawan tersebut dari tabel job_history dimana tanggal dimulai kerjanya karyawan tersebut diurutkan dari yang paling baru.

```
hr=> SELECT first_name, start_date FROM employees INNER JOIN job_history ON employees.employee_id = job_history.employee_id ORDER BY job_history.start_date
```

```
DESC;
```

first_name	start_date
Payam	1999-01-01
Jonathon	1999-01-01
Jonathon	1998-03-24
Den	1998-03-24
Michael	1996-02-17
Jennifer	1994-07-01
Weena	1993-10-28
Lex	1993-01-13
Weena	1989-09-21
Jennifer	1987-09-17

(10 rows)

1. Kesimpulan

Dalam praktikum kali ini Saya mempelajari lebih jauh tentang SQL Shell. Dalam praktikum kali ini Saya mempelajari tentang JOIN antar table dan mempraktekkan ketiga jenis JOIN yaitu, INNER JOIN, LEFT JOIN, dan RIGHT JOIN dengan menggunakan soal yang disediakan.

2. Cek List

No	Elemen Kompetensi	Penyelesaian	
		Selesai	Tidak
1	Perintah atau Command SQL	✓	

3. Form Umpan Balik

Elemen Kompetensi	Waktu Pengerjaan	Kriteria
-------------------	------------------	----------



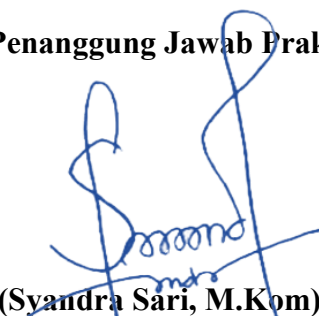
Perintah atau Command SQL	20 Menit	2
---------------------------	----------	---

Kriteria

- 1.Sangat Menarik
- 2.Cukup Menarik
- 3.Kurang Menarik
- 4.Sangat Kurang Menarik

Menyetujui
Jakarta, 1 Juli 2024

Penanggung Jawab Praktikum



((Syandra Sari, M.Kom))

Laboratorium

Kepala



(Is Mardianto, S.Si., M.Kom)

Rekayasa Data

